Report: GOEA

Contents

1 Summary

ABF2 and ABF3 play major role in increased cell division and transcriptional activity as reponse to nitrate stress

- This findings supports the fact, that nitrogen is key signal in cell growth and absorbs nitrate at the root
- The GOEA suggests that ABF2 and ABF3 share major functionality but also divide there tasks, coordinating the response cascade

Shared roles

- Both seem to down regulate the maintenance program of the chromosome and upregulating RNA synthesis
- Both seems to up regulate cell division, RNA programs and down regulating cell shaping processes (actin formation)
- Both acting on ATP regulation
- Both acting on enzymes: Down regulating Isomerase, Dow regulating Nucleases, Up regulating helicases

Different roles

ABF2

- Down regulates Chromatin,

ABF§

- Down regulates Microtubuli, up regulating Peptidase, down regulating environmental stimuli

Cooridnated Roles

- ABF2 seems to stronger act on the down regulation of Heterochromatin, whereas ABF3 acts on the upregulation of euchromatin

2 Control vs ABF2

CC - down

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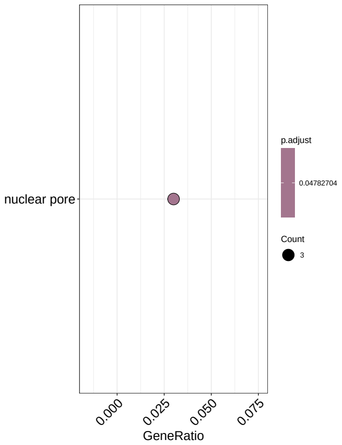

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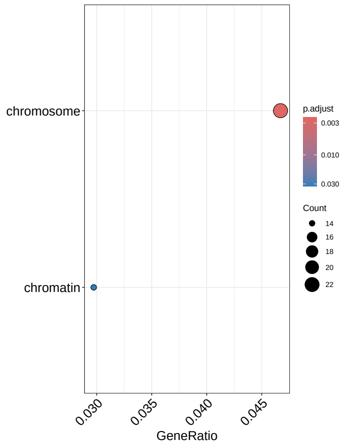

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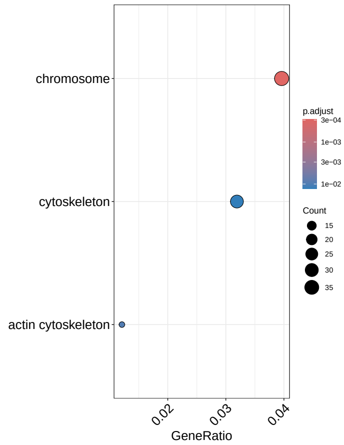

- Downregulation on the Chromsomal level (high significance and GeneRatio) and cytoskeleton level (lower significance and GeneRatio)
- Supporting the idea that DNA specific activity is reduced as well cell shapig

CC - up

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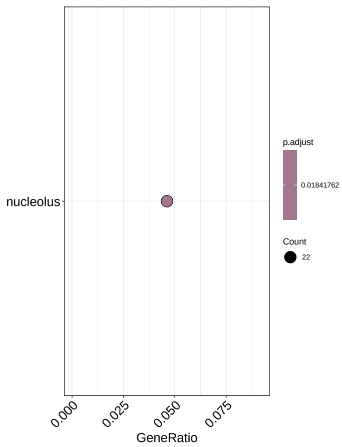

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- Upregulation of the nucleolus term with reasonable significance and GeneRatio
- Suggests an increase in the synthesis of RNA

Interpretation

- Cell's attention moving away from genome maintenance towards the production of RNA molecules

BP - down

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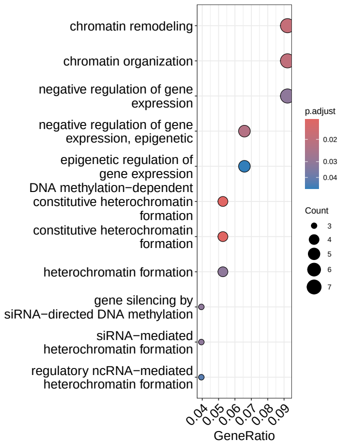

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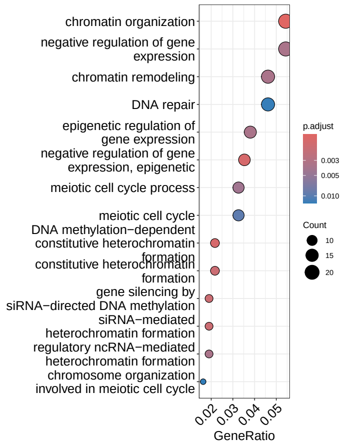

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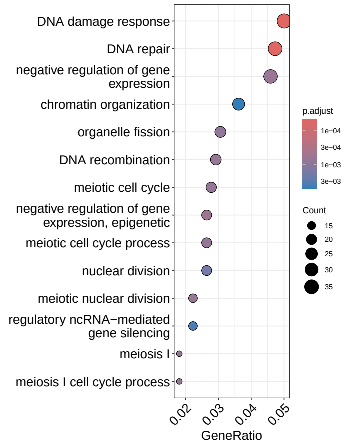

- Downregulation of mostly chromosomal processes like DNA damage response, DNA repair or Chromatin remodeling like heterochromatin formation
- Downregulation of negative regulation of gene expression
- Suggesting that the Chromosome is leaving the state where it is hardly accessible to protein synthesis

BP - up

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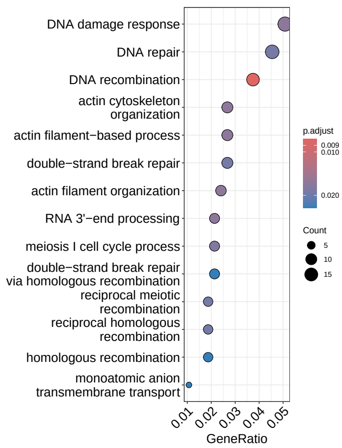

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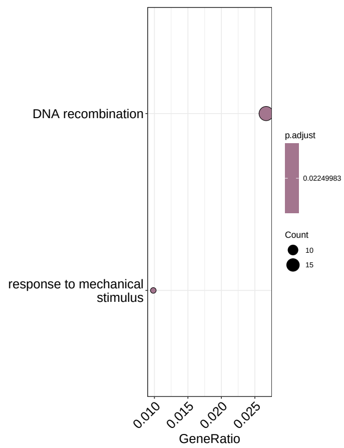

- Upregulation of cell shaping process like actin filament organization or actin cytoskeleton organization (but with lower significance)
- Upregulation o Cell devision terms like DNA recombination or meiosis I cell cycle process
- Upregulation of RNA terms like RNA 3'-end organization

Interpretation

- The biological Process GO terms align with th cellular component GO terms, suggesting that the Cell's going into a state of increased Cell division and transcriptional activity an leaving the maintenance state

MF - down

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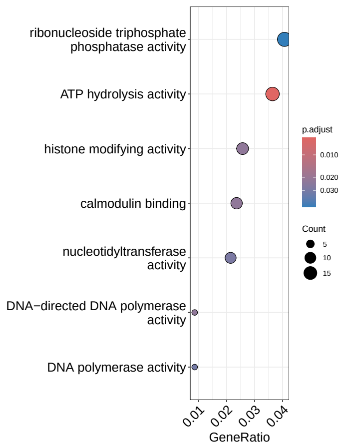

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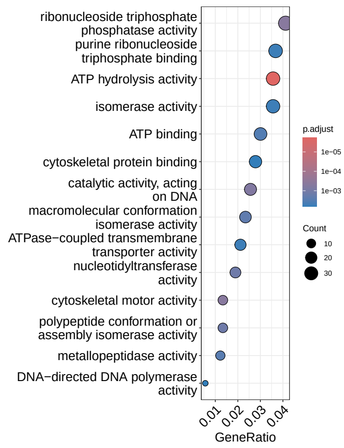

- Downregulation of ATP hydrolysis activity (most significant term) and Polymerase activity
- Both downregulation are a little contradictory to what I would expect
- What aligns is the downregulation of histone modifying activity

MF - up

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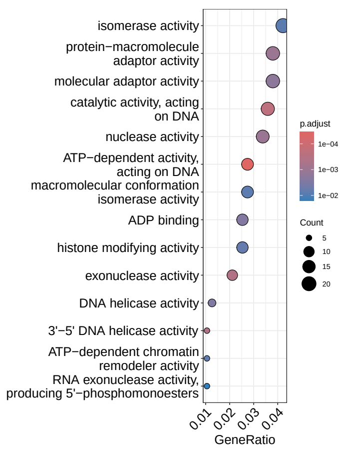

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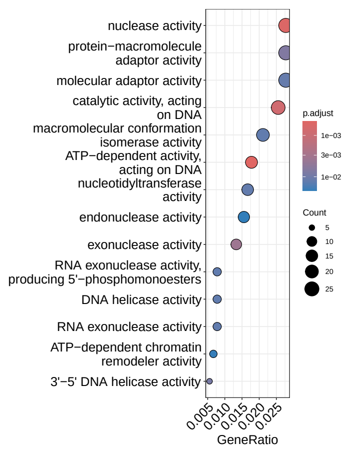

- Upreguated terms mostly allign with the previous interpreations, increased isomerase activity, nuclease activity or ATP-dependent activity acting on DNA and other enzymes

Interpretation

- Since ATP hydrolysis consumes ATP (downregulated) but ATP-dependent activity acting on DNA is upregulated, would could conclude, that the DNA is preserving those ATP molecules from the general housekeeping and moving the energy towards cell division specific tasks

3 Control vs ABF3

CC - down

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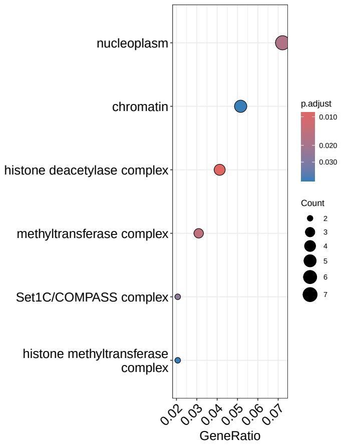

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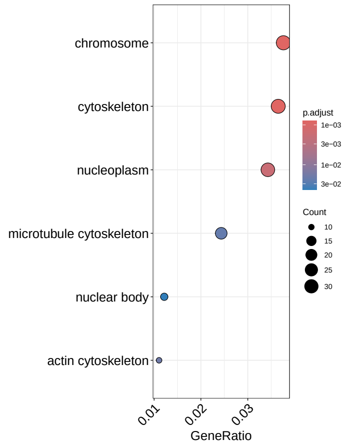
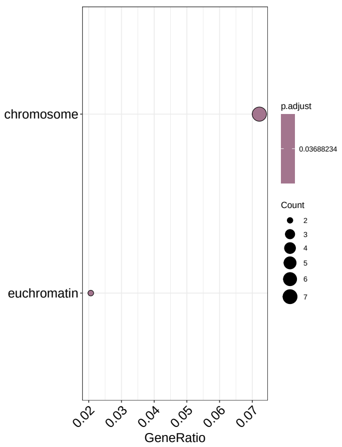


- Detailed GO terms in the top 100 downregulated genes, might allowing a precise localization of the ATF3 activity
- Top 1000 downregulated GO terms similiar to the ABF2 terms, additionally on can see the microtubule cytoskeleton term, again suggestin downregulation of cell shape (or elongation)

CC - up

Top 100



Top 500

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- The localization of upregulation seems not very clear, since only top 100 genes reveal 2 terms, with low significance

Interpretation

- ABF2 had GO terms of downreguation of heterochromatin, now we see that the ABF3 sample seems to be active in euchromatin related tasks, suggesting a activity coordination of both TFs

BP - down

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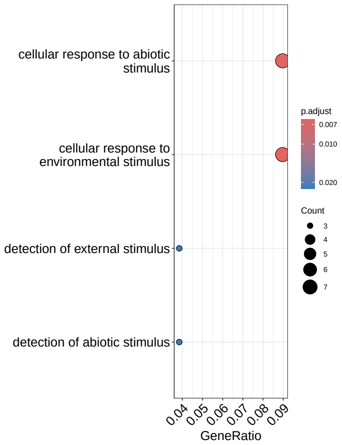

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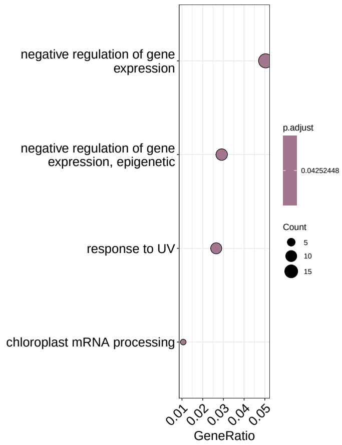

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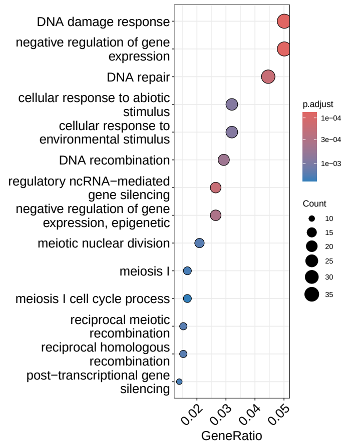


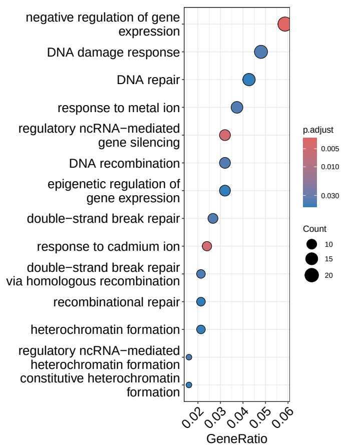
- Similiar to the ABF2 downregulation
- What seems to be outstanding is the cellular response to abiotic and environmental stimulus (high significance and GeneRatio)
- Suggesting that the cell is not in a state looking for stimuli rather in a state of acting on stimuli

BP - up

Top 100

no GO found

Top 500



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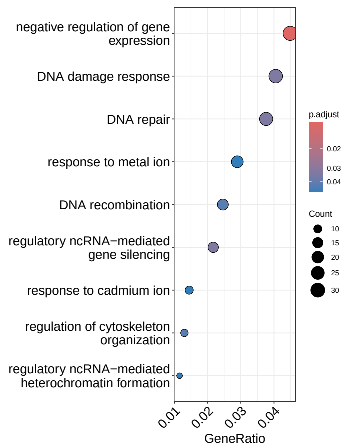

- Almost analog to the ABF2 activity

- Since the term of negative regulation of gene expression appears in both types of regulation, suggesting that there happens a transcriptional porgram shift (downregulation of a maintenance gene cascade towards a division gene cascade)

MF - down

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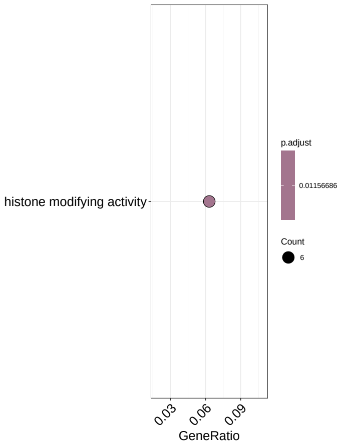

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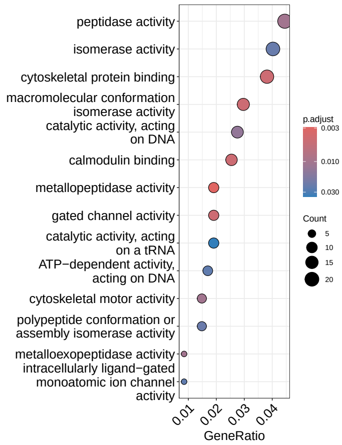

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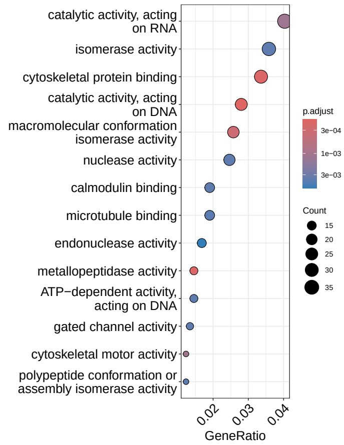

- Analog to the ABF2 with an addtional enzyme: Peptidase

MF - up

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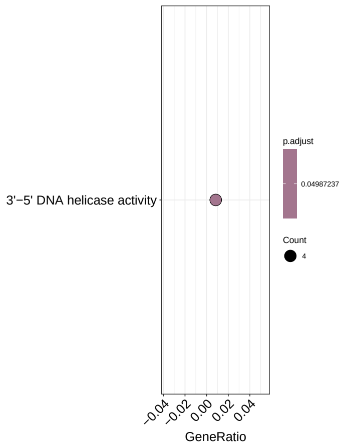

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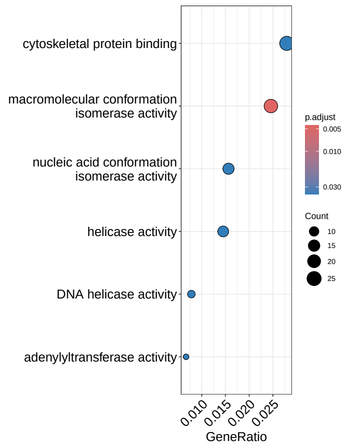

- Also analog two the ABF2 upregulated GO Terms

Interpretation

- Suggest that ABF2 and ABF3 are close players in the upregulation of cell division, where both TFs seem to have slightly different roles especially on what enzymes they act